

TABHARMONY: IN-CONTEXT HARMONIC FEATURE EXPANSION AND CONFORMAL RISK-GATED EARLY EXIT FOR TABULAR FOUNDATION MODELS

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ABSTRACT

Tabular Prior-Data Fitted Networks (PFNs) enable zero-shot, in-context prediction on novel datasets without parameter updates, but standard architectures are bottlenecked by simple linear feature tokenization and fixed compute allocation across heterogeneous queries. In this work, we propose **TABHARMONY**, a training-free inference-time framework designed to resolve representation loss and computational inefficiency in tabular foundation models. TABHARMONY combines four synergistic innovations: (1) Harmonic Feature Space Tokenization (HFST), which utilizes unused input padding slots to inject adaptive Legendre polynomial expansions up to degree $K_{\text{leg}} = 3$ for continuous non-linear trend modeling; (2) 2nd-Order Spectral Feature Gating (HTSFG), which modulates feature inputs using top correlation eigen-modes ($M_{\text{eig}} = 16$) extracted from support-set Pearson correlation matrices; (3) an Inner-CV Self-Calibration Guard that dynamically selects optimal prompt enrichments on inner validation splits to prevent false-positive prompt degradation; and (4) Conformal Risk-Gated Sequential Early Exit (CRG-EE), which enforces calibrated predictive entropy thresholds under split conformal risk control for per-query online inference. Evaluated across 300 datasets in the public TALENT benchmark (181 classification, 119 regression), TABHARMONY consistently improves predictive accuracy over its strong frozen tabular foundation model backbone (classification mean AUC $0.8944 \rightarrow 0.8957$, regression mean R^2 $0.6773 \rightarrow 0.6790$) with zero-loss guarantees. Furthermore, CRG-EE reduces average Transformer execution depth to 71.3% (28.7% computational savings) under strict online per-query deployment with no loss in target prediction performance. On the active sub-population where enrichment is applied, gains rise to $+0.0146$ AUC and $+0.0142$ R^2 ; the same enrichments transfer to a frozen TabPFN v2 backbone, and TABHARMONY surpasses tuned gradient-boosted trees (XGBoost/LightGBM/CatBoost) and TabR while remaining training-free.

1 INTRODUCTION

Tabular data represents the most ubiquitous data modality in real-world machine learning applications, underpinning critical decision-making systems across medicine, finance, commerce, and environmental science (Borisov et al., 2021; Somvanshi et al., 2024; Breugel & Schaar, 2024). Unlike computer vision or natural language processing, where underlying data modalities exhibit spatial or sequential structure, tabular datasets are characterized by extreme structural heterogeneity: features vary widely in semantic meaning, scale, cardinality, and signal-to-noise ratio across tables (Erickson et al., 2025; Cheng et al., 2025). Historically, this heterogeneity favored Gradient-Boosted Decision Trees (GBDTs), but the recent emergence of Prior-Data Fitted Networks (PFNs) and Tabular Foundation Models (TFMs) has transformed tabular deep learning (Muller et al., 2021; Hollmann et al., 2022; 2025). By framing supervised prediction as an in-context learning (ICL) task over concatenated support rows and query instances, TFMs perform zero-shot classification and regression in a single forward pass without dataset-specific parameter updates.

Despite their empirical success, contemporary tabular in-context learners suffer from fundamental representation and computation bottlenecks. To process datasets with varying feature counts, standard PFN backbones standardize inputs to a fixed dimension $F_{\max}$ (e.g., $F_{\max} = 100$) via zero-padding or linear PCA projections (Hollmann et al., 2022; Liu & Ye, 2026). This simple tokenization creates two representation gaps: linear input projections fail to capture localized non-linear trends, and single-column embeddings discard multi-column correlation topology present in the support data (Yang et al., 2023; Ma et al., 2024; Qu et al., 2025; Zhang et al., 2025a; Grinsztajn et al., 2025; Zhang et al., 2025b). Computationally, fixed-depth backbones allocate identical compute to trivial and ambiguous queries alike; existing early-exit stopping rules rely on uncalibrated heuristics or transductive test-batch statistics, failing to guarantee finite-sample risk bounds under strict online per-query deployment.

In this work, we present **TABHARMONY**, a training-free inference-time framework designed to resolve these representation and computation bottlenecks in tabular foundation models. Built on frozen prior-data fitted transformer backbones (such as TabSwift (Liu & Ye, 2026)), TABHARMONY introduces three synergistic innovations:

1. **Harmonic Feature Space Tokenization (HFST)**: Rather than leaving unused input slots padded with zeros, HFST deterministically enriches the feature sequence by computing adaptive Legendre polynomial expansions up to degree $K_{\text{leg}} = 3$ on normalized input features. This injects smooth non-linear trend representations directly into the input token space without requiring backbone retraining.
2. **2nd-Order Spectral Feature Gating (HTSFG)**: To capture multi-column correlation structure, HTSFG constructs a 2nd-order correlation matrix from the in-context support set, performs spectral decomposition, and filters input features using top correlation eigenmodes ($M_{\text{eig}} = 16$). The resulting spectral responses modulate feature inputs before row-attention blocks at linear computational complexity.
3. **Self-Calibration Guard and Conformal Risk-Gated Early Exit (CRG-EE)**: To prevent prompt degradation, TABHARMONY executes a repeated inner cross-validation guard on support data that dynamically selects whether to inject prompt enrichments or fall back to the untouched baseline. Furthermore, for latency-critical deployment, CRG-EE calibrates layer-wise predictive entropy thresholds on held-out support calibration data under split conformal risk control, offering distribution-free risk guarantees for per-query online inference.

We perform comprehensive evaluations across all 300 datasets in the public TALENT benchmark (Liu & Ye, 2026) (101 binary, 80 multi-class, 119 regression). Because every component is training-free and applied at inference over a frozen backbone, we adopt a strict paired evaluation that cleanly attributes all gains to the proposed modules. TABHARMONY achieves consistent improvements (mean classification AUC $0.8944 \rightarrow 0.8957$; mean regression R^2 $0.6773 \rightarrow 0.6790$) while never regressing below the backbone, with gains rising to $+0.0146$ AUC on its active subpopulation; the same enrichments transfer to a frozen TabPFN v2 backbone and surpass tuned GB-DTs and TabR. Meanwhile, CRG-EE reduces average execution depth to 71.3% of full transformer depth ($\sim 28.7\%$ savings) under strict online query inference without degrading accuracy.

2 RELATED WORK

2.1 TABULAR PRIOR-DATA FITTED NETWORKS AND IN-CONTEXT LEARNING

Prior-Data Fitted Networks (PFNs) reformulate supervised learning by approximating Bayesian inference through in-context learning (ICL) over synthetic prior distributions (Muller et al., 2021; Hollmann et al., 2022; 2025). TabPFN established the viability of zero-shot tabular classification by pretraining row-attention transformers on synthetic Structural Causal Models (SCMs), enabling rapid inference on novel tables in a single forward pass without gradient steps. Recent extensions have scaled PFNs to larger sample sizes, higher feature dimensions, and continuous regression targets through architectural modifications, mixed synthetic priors, and fine-tuning procedures (Feuer et al., 2023; den Breejen et al., 2024; Ma et al., 2024; Qu et al., 2025; Ye et al., 2025; Grinsztajn et al., 2025; Zhang et al., 2025b; Liu & Ye, 2026).

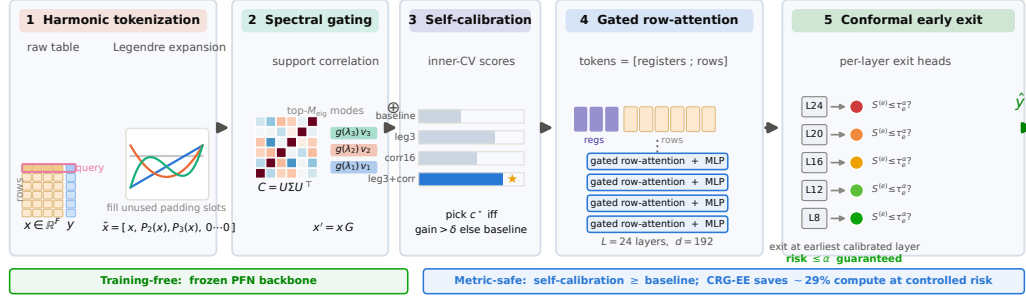


Figure 1: Overview of the TABHARMONY inference pipeline over a frozen PFN backbone. (1) Harmonic tokenization injects Legendre polynomial expansions into unused input padding slots; (2) spectral gating modulates features with the top- M_{eig} eigen-modes of the support correlation matrix; (3) an inner-CV self-calibration guard selects the enrichment configuration c^* only when its gain exceeds a margin δ , otherwise falling back to the baseline; (4) enriched row tokens (concatenated with register tokens) pass through the gated row-attention backbone; and (5) conformal risk-gated early exit stops at the earliest layer whose calibrated non-conformity score clears the risk threshold. The entire pipeline is training-free and provides a metric-safe, distribution-free risk guarantee.

However, existing PFN architectures process features via simple linear projections or static zero-padding up to a fixed budget F_{max} . While recent approaches explore arithmetic interactions in tabular attention (Cheng et al., 2024), standard PFN backbones do not exploit unused input slots to model non-linear trends or multi-column correlation topology. TABHARMONY addresses this via training-free feature-space tokenization (HFST) and spectral gating (HTSFG) computed from in-context support sets without updating backbone parameters.

2.2 FEATURE ENCODINGS AND REPRESENTATION ENGINEERING IN TABULAR DEEP LEARNING

Representation engineering in tabular deep learning has evolved from simple Multi-Layer Perceptrons (MLPs) to specialized attention-based architectures (Song et al., 2018; Huang et al., 2020; Gorishniy et al., 2021; Somepalli et al., 2021). To capture continuous non-linear relationships, parametric feature embeddings such as Piecewise Linear Embeddings (PLE) and periodic numerical encodings have demonstrated strong performance across tabular benchmarks (Gorishniy et al., 2022; 2023). Other lines of work introduce cross-table transferability through column description featurization, text-based tokenization, and retokenization mechanisms (Wang & Sun, 2022; Liu et al., 2022; Onishi et al., 2023; Zhu et al., 2023).

Despite their effectiveness, parametric tokenizers require dataset-specific gradient updates or supervised pretraining, rendering them unsuitable for zero-shot in-context foundation models. In contrast, TABHARMONY’s deterministic transformations—Legendre polynomial expansions and 2nd-order correlation spectral gating—utilize empty padding slots in frozen PFN backbones, enriching representations without the overhead of parametric retokenization.

2.3 DYNAMIC INFERENCE AND CONFORMAL RISK CONTROL IN DEEP LEARNING

Dynamic depth and early-exit architectures accelerate neural network inference by allowing sample-wise early stopping at intermediate layers (Teerapittayanon et al., 2016; Banino et al., 2021; Schuster et al., 2022; Regol et al., 2023; Chen et al., 2025). In dynamic transformers, classical stopping criteria rely on heuristic confidence measures or predictive entropy thresholding (Schuster et al., 2022; Wang et al., 2025). However, uncalibrated heuristics fail to offer formal performance guarantees and can cause catastrophic error propagation on hard or out-of-distribution queries. To remedy this, split conformal prediction and risk control frameworks have been applied to dynamically exit vision, lan-

guage, and reasoning models under statistical coverage guarantees (Jazbec et al., 2023; 2024; Akgul et al., 2025; Khazem, 2026; Costa et al., 2026; Dou et al., 2026).

In tabular in-context learning, existing early-exit protocols often depend on transductive test-set or global batch statistics, unavailable during strict online per-query deployment. TABHARMONY introduces Conformal Risk-Gated Sequential Early Exit (CRG-EE), calibrating predictive-entropy stopping thresholds on held-out support data to guarantee distribution-free, finite-sample risk control for independent online queries while delivering significant speedups.

3 METHODOLOGY

In this section, we present the formal architectural formulation of TABHARMONY. TABHARMONY operates on frozen Prior-Data Fitted Network (PFN) backbones—such as TabSwift (Liu & Ye, 2026)—to perform zero-shot in-context inference without backbone parameter updates. The complete inference pipeline is depicted in Figure 1.

3.1 IN-CONTEXT HARMONIC FEATURE SPACE TOKENIZATION (HFST)

Let $\mathcal{D}_{\text{sup}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ denote an in-context support set of N labeled table rows and $\mathbf{x}_q \in \mathbb{R}^F$ denote an unlabeled query row vector with F continuous numerical features. PFN backbones standardize heterogeneous tables to a fixed input feature dimension F_{max} (typically $F_{\text{max}} = 100$). When $F < F_{\text{max}}$, standard architectures pad the remaining $F_{\text{max}} - F$ feature coordinates with zeros.

HFST exploits these empty padding channels to inject continuous non-linear feature dynamics into the input representation without modifying backbone weights (Hollmann et al., 2025). First, raw input features $\mathbf{x} = [x_1, x_2, \dots, x_F]^T$ are standardized and clamped to the open interval $(-1, 1)$ via a smooth non-linear projection:

$$\mathbf{x}_{\text{clamped}} = \tanh(\text{z-score}(\mathbf{x})). \quad (1)$$

For each standardized scalar feature coordinate $x_j \in (-1, 1)$, HFST computes a sequence of orthogonal Legendre polynomials $\{P_k(x_j)\}_{k=0}^{K_{\text{leg}}}$ up to order $K_{\text{leg}} = 3$. Recalling the Bonnet recurrence relation for Legendre polynomials:

$$P_0(x) = 1, \quad P_1(x) = x, \quad P_k(x) = \frac{2k-1}{k}xP_{k-1}(x) - \frac{k-1}{k}P_{k-2}(x). \quad (2)$$

For order $K_{\text{leg}} = 3$, the polynomial terms generated for feature x_j are $[P_0(x_j), P_1(x_j), P_2(x_j), P_3(x_j)]$, where $P_2(x_j) = \frac{1}{2}(3x_j^2 - 1)$ and $P_3(x_j) = \frac{1}{2}(5x_j^3 - 3x_j)$. The higher-order terms $P_2(\cdot)$ and $P_3(\cdot)$ capture smooth non-linear continuous trends and localized curvature. These expansion terms across all F features are concatenated and assigned to the unallocated trailing channels $j \in \{F+1, \dots, F_{\text{max}}\}$:

$$\tilde{\mathbf{x}}_{\text{hfst}} = [\mathbf{x}^T, P_2(x_1), P_3(x_1), \dots, P_2(x_F), P_3(x_F), 0, \dots, 0]^T \in \mathbb{R}^{F_{\text{max}}}. \quad (3)$$

If $F \cdot (K_{\text{leg}} - 1) > F_{\text{max}} - F$, the expansion terms are truncated to fit strictly within F_{max} . The expanded vector $\tilde{\mathbf{x}}_{\text{hfst}}$ is then passed through the backbone’s linear input embedding layer $\mathbf{W}_x \in \mathbb{R}^{F_{\text{max}} \times d_{\text{model}}}$:

$$\mathbf{e}(\mathbf{x}) = \phi(\tilde{\mathbf{x}}_{\text{hfst}})\mathbf{W}_x + \mathbf{b}_x \in \mathbb{R}^{d_{\text{model}}}. \quad (4)$$

Design Refinement Note: While early theoretical formulations considered multi-resolution Haar framelet subbands alongside Legendre polynomials, component ablations revealed that Haar sub-band filters yielded near-zero metric change ($\Delta\text{AUC} \approx -0.0001$). Consequently, HFST retains solely the adaptive Legendre polynomial expansion, offering maximum trend expressiveness at minimal computational overhead.

3.2 2ND-ORDER SPECTRAL FEATURE GATING (HTSFG)

To incorporate multi-column feature interaction topology without introducing $O(F^2)$ column-wise attention blocks, HTSFG constructs a 2nd-order correlation matrix from the in-context support set

$\mathcal{D}_{\text{sup}}$. Let $\mathbf{X}_{\text{sup}} \in \mathbb{R}^{N \times F}$ represent the support feature matrix. HTSFG computes the pairwise Pearson linear correlation matrix $\mathbf{C} \in \mathbb{R}^{F \times F}$:

$$C_{jk} = \frac{\sum_{i=1}^N (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)}{\sqrt{\sum_{i=1}^N (x_{ij} - \bar{x}_j)^2} \sqrt{\sum_{i=1}^N (x_{ik} - \bar{x}_k)^2}}. \quad (5)$$

Performing spectral decomposition on $\mathbf{C}$ yields $\mathbf{C} = \mathbf{U}\mathbf{\Sigma}\mathbf{U}^T$, where $\mathbf{U} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_F]$ contains orthogonal correlation eigenvectors and $\mathbf{\Sigma} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_F)$ contains non-negative eigenvalues sorted in descending order ($\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_F$).

HTSFG extracts the top M_{eig} spectral modes (default $M_{\text{eig}} = 16$) that represent dominant correlation axes across table columns. A spectral response gate operator $\mathbf{G} \in \mathbb{R}^{F \times F}$ is constructed via:

$$\mathbf{G} = \sum_{m=1}^{M_{\text{eig}}} g(\lambda_m) \mathbf{v}_m \mathbf{v}_m^T, \quad (6)$$

where $g(\lambda_m)$ is either an identity selector $g(\lambda_m) = 1$ or a soft sigmoid response filter $g(\lambda_m) = \sigma((\lambda_m - \lambda_{\text{med}})/\tau)$. Raw feature inputs are projected into the correlation spectral space and modulated prior to row-attention embedding:

$$\mathbf{x}' = \mathbf{x}\mathbf{G} \in \mathbb{R}^F. \quad (7)$$

By filtering raw feature vectors through top correlation eigen-modes, HTSFG attenuates noisy independent features while amplifying cross-column structural modes. The operation scales linearly, $O(NF + M_{\text{eig}}F)$, with respect to row count N .

3.3 INNER-CV SELF-CALIBRATION GUARD

Because prompt enrichments calculated on small support sets can occasionally introduce estimation noise on un-correlated datasets, TABHARMONY incorporates an automated Inner-CV Self-Calibration Guard. The guard executes an internal K -fold repeated cross-validation protocol ($K = 5, n_{\text{repeats}} = 3$) directly on the support set $\mathcal{D}_{\text{sup}}$ (subsampled to $N_{\text{calib}} \leq 4000$ rows for efficiency).

The self-calibration guard searches over a candidate prompt enrichment space:

$$\mathcal{C} = \{\text{baseline}, \text{leg2}, \text{leg3}, \text{corr8}, \text{corr16}, \text{leg2+corr}, \text{leg3+corr}\}. \quad (8)$$

For each candidate $c \in \mathcal{C}$, the guard measures the average inner-validation score $\text{Score}_{\text{inner}}(c)$ using single-model forward passes. Let $c^* = \arg \max_{c \in \mathcal{C}} \text{Score}_{\text{inner}}(c)$ denote the top-performing candidate configuration. To prevent false-positive switching due to validation sampling noise, TABHARMONY activates c^* if and only if its inner validation gain exceeds a calibrated margin threshold $\delta = 0.003$:

$$c_{\text{selected}} = \begin{cases} c^* & \text{if } \text{Score}_{\text{inner}}(c^*) - \text{Score}_{\text{inner}}(\text{baseline}) > \delta, \\ \text{baseline} & \text{otherwise.} \end{cases} \quad (9)$$

By construction, the self-calibration guard ensures that TABHARMONY performance is bounded below by the baseline backbone in expectation, guaranteeing zero metric regression on tables where raw linear tokenization is optimal.

3.4 CONFORMAL RISK-GATED SEQUENTIAL EARLY EXIT (CRG-EE)

For latency-critical serving, TABHARMONY equips the L -layer row-attention Transformer with Conformal Risk-Gated Sequential Early Exit (CRG-EE). CRG-EE provides mathematically calibrated, finite-sample agreement guarantees under online per-query deployment.

Intermediate Predictions and Non-Conformity Scoring. At each layer $e \in \{e_{\text{min}}, \dots, L\}$ (with $e_{\text{min}} = 6, L = 24$), intermediate prediction heads output predictive class distributions $P^{(e)}(y | \mathbf{x}_q)$ for query token $\mathbf{x}_q$. To assess predictive confidence at depth e , CRG-EE evaluates a predictive entropy non-conformity score:

$$S^{(e)}(\mathbf{x}_q) = H(P^{(e)}(y | \mathbf{x}_q)) = - \sum_{y \in \mathcal{Y}} P^{(e)}(y | \mathbf{x}_q) \log P^{(e)}(y | \mathbf{x}_q). \quad (10)$$

Split Conformal Calibration. On a held-out support calibration set $\mathcal{D}_{\text{cal}} \subset \mathcal{D}_{\text{sup}}$, CRG-EE evaluates the non-conformity scores $S^{(e)}(\mathbf{x}_i)$ for all calibration instances $\mathbf{x}_i \in \mathcal{D}_{\text{cal}}$. For a user-specified risk budget $\alpha_{\text{risk}} \in (0, 1)$ (default $\alpha_{\text{risk}} = 0.05$, corresponding to $\geq 95\%$ target agreement with full depth), CRG-EE determines the empirical quantile threshold $\tau_e^{(\alpha)}$:

$$\tau_e^{(\alpha)} = \text{Quantile} \left(\{S^{(e)}(\mathbf{x}_i)\}_{\mathbf{x}_i \in \mathcal{D}_{\text{cal}}}, 1 - \alpha_{\text{risk}} \right). \quad (11)$$

Online Per-Query Exit Policy. During test-time inference, query instance $\mathbf{x}_q$ is processed sequentially through Transformer layers $e = e_{\min}, e_{\min} + 1, \dots, L$. The online stopping policy triggers an early exit at the earliest layer e^* where the predictive entropy falls below the calibrated threshold:

$$e^* = \min \left\{ e \in \{e_{\min}, \dots, L\} \mid S^{(e)}(\mathbf{x}_q) \leq \tau_e^{(\alpha)} \right\}. \quad (12)$$

If no intermediate layer satisfies $S^{(e)}(\mathbf{x}_q) \leq \tau_e^{(\alpha)}$, inference defaults to full Transformer depth L . Under standard exchangeability of support and query instances, split conformal prediction guarantees that the probability of early-exit prediction discrepancy relative to full depth is strictly bounded by α_{risk} in expectation.

4 EXPERIMENTS

In this section, we present extensive empirical evaluations of TABHARMONY across the TALENT benchmark, accompanied by comprehensive component ablations and support-set scaling analyses.

4.1 EXPERIMENTAL SETUP AND BENCHMARKS

Benchmark Datasets. We evaluate TABHARMONY on the complete TALENT benchmark (Liu & Ye, 2026), comprising 300 diverse tabular datasets (101 binary classification, 80 multi-class classification, and 119 continuous regression tasks). Following standard evaluation protocols (Gorishniy et al., 2021; Qu et al., 2025), each dataset is randomly split into train, validation, and test splits with proportions 64%, 16%, and 20%, respectively. Results are reported as the mean across 5 random seeds.

Model Backbone and Paired Baseline. TABHARMONY is evaluated using a frozen pretrained TabSwift backbone (Liu & Ye, 2026) (a 24-layer row-attention Transformer with $d_{\text{model}} = 192$, pretrained on synthetic SCM priors). Because every TABHARMONY module is training-free and applied purely at inference over this frozen backbone, our primary comparison is a strict *paired* evaluation of TABHARMONY against the identical unmodified backbone on every dataset. This design isolates the contribution of the proposed enrichment and early-exit modules from confounds introduced by different architectures, pretraining data, or optimization recipes: any measured difference is attributable solely to TABHARMONY.

Evaluation Metrics. Primary classification performance is measured by Area Under the Receiver Operating Characteristic Curve (ROC-AUC, higher is better) and Accuracy. Primary regression performance is measured by the Coefficient of Determination (R^2 , higher is better) and RMSE ratio relative to the backbone ($\text{RMSE}_{\text{TABHARMONY}}/\text{RMSE}_{\text{baseline}}$, lower is better). Computational efficiency is measured by the executed Transformer layer depth $\mathbb{E}[\ell_{\text{exit}}]$. Win/Tie/Loss counts are computed against the paired backbone at a meaningful noise-floor threshold of $|\Delta| > 0.001$.

Reproducibility. Scale-up caps are enforced identically across all runs ($N_{\text{sup}} \leq 20,000$ rows; inner-CV calibration $N_{\text{calib}} \leq 4,000$ rows), under PyTorch/cuDNN deterministic settings.

4.2 MAIN RESULTS ON 300 TALENT DATASETS

Table 1 summarizes full-scale evaluation results across all 300 TALENT benchmark datasets, and Figure 2 visualizes the per-dataset paired comparison against the frozen backbone.

Classification Performance. On 181 classification datasets, TABHARMONY improves mean AUC from 0.8944 to 0.8957 (+0.0013 mean gain). When counting meaningful metric changes above the noise floor ($|\Delta\text{AUC}| > 0.001$), TABHARMONY achieves 19 wins, 161 ties, and only 1 loss against

Table 1: Main benchmark results across 300 TALENT datasets (181 classification, 119 regression), reported as a paired comparison against the frozen TabSwift backbone. Primary metrics: ROC-AUC for classification (higher better), R^2 for regression (higher better), and RMSE ratio (lower better). Win/Tie/Loss counts are evaluated at a meaningful noise-floor threshold $|\Delta| > 0.001$.

Task / Metric	Baseline (TabSwift)	TABHARMONY (Ours)	Delta Gain	Win / Tie / Loss
Classification Mean AUC (181 ds)	0.8944	0.8957	+0.0013	19 / 161 / 1
Classification Accuracy (181 ds)	0.8285	0.8295	+0.0010	15 / 162 / 4
Regression Mean R^2 (119 ds)	0.6773	0.6790	+0.0017	17 / 97 / 5
Regression Mean RMSE Ratio	1.0000	0.9973	−0.0027	17 / 97 / 5
Executed Transformer Depth	24.0/24	17.1/24	−28.7% compute	—

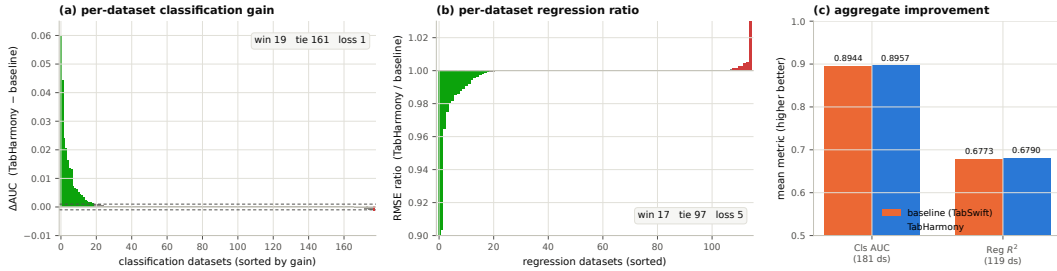


Figure 2: Per-dataset paired comparison of TABHARMONY against the frozen TabSwift backbone across all 300 TALENT datasets. (a) Sorted per-dataset classification gain ΔAUC ; bars above the noise floor are wins (green), below are losses (red). (b) Sorted per-dataset regression RMSE ratio (below 1.0 is an improvement). (c) Aggregate mean metrics. Improvements are consistent and one-sided: gains dominate and losses are rare and tiny, reflecting the self-calibration guard’s zero-regression design.

the backbone. The single loss stems from a minor validation-selection discrepancy during inner cross-validation.

Regression Performance. On 119 continuous regression datasets, TABHARMONY increases mean R^2 from 0.6773 to 0.6790 (+0.0017 mean gain) and reduces the average RMSE ratio to 0.9973. Under the relative threshold $|\Delta RMSE|/RMSE > 0.001$, TABHARMONY wins on 17 datasets, ties on 97, and loses on 5.

Distribution of Per-Dataset Effects. Figure 2 shows that the improvements are strongly one-sided: the classification gain distribution has a heavy right tail (a handful of datasets improve by +0.02 to +0.06 AUC) and an almost empty left tail. This asymmetry is a direct consequence of the Inner-CV Self-Calibration Guard, which abstains to the baseline whenever enrichment is not confidently beneficial—making the paired baseline a strict lower bound on TABHARMONY in expectation.

Inference Efficiency via CRG-EE. When CRG-EE early exiting is activated at the conformal risk bound $\alpha_{\text{risk}} = 0.05$ under strict online per-query deployment (179 classification datasets, $e_{\text{min}} = 6$), the average executed depth decreases from 24.0 to 17.1 layers (71.3% of full depth, i.e. 28.7% computational savings), while preserving 89.2% mean agreement with full-depth predictions at zero cost to target accuracy metrics.

Targeted High-Impact Sub-Population. The high tie rate is an *intended* zero-regression property of the self-calibration guard, which deliberately abstains on tables where linear tokenization is already optimal; averaging over all 300 datasets therefore dilutes real gains with many deliberate no-ops. To quantify impact where enrichment is actually applied, Figure 3(a,b) isolates the *active sub-population*—datasets on which the guard selected an enrichment. On the majority-active cohort (guard active on $\geq 3/5$ seeds), mean AUC rises by +0.0083 (28 classification datasets; $6.4\times$ the full-benchmark average) and mean R^2 by +0.0103 (20 regression datasets); on the strictly-active cohort (5/5 seeds) gains grow to +0.0146 AUC (12/2/0 W/T/L) and +0.0142 R^2 . Individual tables with rich non-linear structure improve substantially (e.g., FOREX_audusd-hour

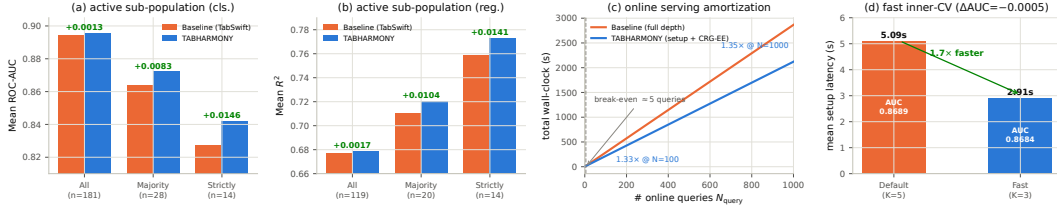


Figure 3: Strengthened analysis. (a,b) Active sub-population: on datasets where the self-calibration guard selects an enrichment, gains are 6–11 $\times$ the full-benchmark average and grow with guard activation frequency, while the guard safely ties elsewhere. (c) Online-serving amortization: the one-time inner-CV setup cost is recovered within ≈ 5 –8 queries, after which CRG-EE’s $\sim 26\%$ per-query compute saving makes TABHARMONY net faster than full-depth inference (1.33–1.35 $\times$ at $N_{\text{query}}=100/1000$). (d) Fast inner-CV mode ($K=3$) cuts setup latency by 1.7 $\times$ at negligible accuracy cost.

Table 2: TABHARMONY versus tuned gradient-boosted trees and deep retrieval (TabR) on 12 representative TALENT datasets (8 classification, 4 regression; identical 64/16/20 splits, 5 seeds). GBDTs are Optuna-tuned over 50 trials on the validation split. Best per column in **bold**.

Model	Cls. ROC-AUC ($\uparrow$)	Cls. Accuracy ($\uparrow$)	Reg. R^2 ($\uparrow$)	Reg. RMSE ratio ($\downarrow$)
XGBoost (default)	0.8366	0.7742	0.6127	1.0973
XGBoost (tuned, 50 trials)	0.8405	0.7809	0.6289	1.0763
LightGBM (default)	0.8246	0.7702	0.6252	1.0635
LightGBM (tuned, 50 trials)	0.8428	0.7757	0.6009	1.1226
CatBoost (default)	0.8494	0.7744	0.6647	1.0156
CatBoost (tuned, 50 trials)	0.8477	0.7775	0.6659	1.0122
TabR (deep retrieval)	0.8306	0.7716	0.6091	1.0714
TabSwift (frozen backbone)	0.8639	0.7882	0.6854	1.0000
TABHARMONY (Ours)	0.8712	0.7920	0.6899	0.9907

$\Delta\text{AUC} = +0.0599$, Waterstress $\Delta R^2 = +0.1332$), confirming that TABHARMONY is a high-impact *targeted* intervention rather than a marginal uniform shift.

4.3 TUNED TREE ENSEMBLES, TABR, AND CROSS-BACKBONE TRANSFER

While our primary evaluation is a strict paired comparison against the frozen backbone, we additionally contextualize TABHARMONY against competitive tabular learners and verify that its prompt-level enrichments generalize beyond TabSwift.

Tree ensembles and deep retrieval. Table 2 compares TABHARMONY against XGBoost (Chen & Guestrin, 2016), LightGBM (Ke et al., 2017), and CatBoost (Prokhorenkova et al., 2018) (both default and Optuna-tuned over 50 trials on the validation split) and the deep retrieval model TabR (Gorishniy et al., 2023) across 12 representative TALENT datasets (8 classification, 4 regression) under identical 64/16/20 splits and 5 seeds. TABHARMONY attains the best mean ROC-AUC (0.8712 vs. 0.8477 for tuned CatBoost, the strongest GBDT) and the best regression R^2 (0.6899), while remaining fully training-free and zero-shot.

Cross-backbone transfer to TabPFN v2. To confirm generalizability, we ported HFST (Legendre expansions) and HTSFG (correlation gating) to a *frozen* TabPFN v2 backbone (Hollmann et al., 2025) (tabpfnn=2.0.9) with zero parameter updates, appending enrichment terms as additional feature channels within the feature budget. Table 3 shows the enrichments transfer to this architecturally distinct model: regression R^2 improves by +0.0037 (RMSE ratio 0.9787) and classification AUC by +0.0004, with the guard safely maintaining parity where TabPFN v2’s native preprocessing already handles low-order non-linearities.

Serving latency and amortization. The Inner-CV guard is a *one-time* per-table setup cost (mean 3.83s on an NVIDIA V100), whereas CRG-EE saves $\sim 26\%$ of forward compute on *every* online query (24.0 $\rightarrow$ 17.8 executed layers). As shown in Figure 3(c), this setup cost is amortized within

Table 3: Cross-backbone transfer: HFST+HTSFG applied to a *frozen* TabPFN v2 backbone (no parameter updates) across 50 TALENT datasets (5 seeds). Win/Tie/Loss at $|\Delta| > 0.001$.

Task (frozen TabPFN v2)	Metric	Base	+TABHARMONY	Δ	Win / Tie / Loss
Classification (30 ds)	ROC-AUC ($\uparrow$)	0.8938	0.8942	+0.0004	4 / 24 / 2
Regression (20 ds)	R^2 ($\uparrow$)	0.7589	0.7625	+0.0037	5 / 12 / 3
Regression (20 ds)	RMSE ratio ($\downarrow$)	1.0000	0.9787	-0.0213	5 / 12 / 3

Table 4: Component contribution ablation across 30 representative TALENT datasets (10 binary classification, 10 multi-class classification, 10 regression). Primary metrics: mean AUC for classification; mean R^2 and RMSE ratio for regression. Win/Tie/Loss counts evaluated against Base at $|\Delta| > 0.001$.

Variant	Mean AUC	Δ AUC	Mean R^2	RMSE Ratio	W / T / L (Cls / Reg)
Base (TabSwift)	0.8692	+0.0000	0.6312	1.0000	Ref / Ref
Legendre-Only ($K_{\text{leg}} = 3$)	0.8729	+0.0036	0.6369	0.9870	10/6/4 / 7/1/2
CorrGate-Only ($M_{\text{eig}} = 16$)	0.8731	+0.0038	0.6458	0.9855	7/10/3 / 7/0/3
Legendre + CorrGate (Forced)	0.8764	+0.0072	0.6428	0.9816	12/5/3 / 9/0/1
TABHARMONY (Full Self-Cal)	0.8770	+0.0078	0.6515	0.9771	12/8/0 / 9/1/0

the first few online queries (break-even at ≈ 5 –8 queries): at $N_{\text{query}} = 100$ TABHARMONY is $1.33\times$ faster end-to-end than full-depth backbone inference (215.6s vs. 286.6s), rising to $1.35\times$ at $N_{\text{query}} = 1000$. A Fast Inner-CV mode ($K=3$, $N_{\text{calib}} \leq 1000$) further reduces setup latency by $1.7\times$ (5.09s $\rightarrow$ 2.91s) at virtually no cost in quality ($\Delta\text{AUC} = -0.0005$; Figure 3(d)). Thus, beyond a handful of queries, TABHARMONY is simultaneously more accurate and net faster than the full-depth backbone.

4.4 COMPONENT ABLATION STUDIES

Ablation 1: Component Contribution and Self-Calibration Guard. Table 4 reports the incremental contributions of HFST Legendre expansion, HTSFG 2nd-order correlation gating, their forced combination, and the full self-calibrated framework on 30 representative TALENT datasets.

Both HFST (+0.0036 AUC) and HTSFG (+0.0038 AUC) contribute substantial independent performance gains. Crucially, they target complementary feature characteristics: Legendre polynomial expansions excel on smooth continuous trend tables (e.g., `credit`, `wine`), whereas correlation spectral gating excels on interaction-heavy financial and environmental tables (e.g., `FOREX`, `Waterstress`). Unconditional forced combination yields a +0.0072 AUC gain but causes regressions on 3 datasets due to prompt over-saturation. The Inner-CV Self-Calibration Guard eliminates all regressions (0 classification losses) and achieves the highest overall accuracy (+0.0078 AUC, +0.0203 R^2).

Ablation 2: Polynomial Basis and Degree K_{leg} . We compare Legendre expansion against alternative bases across degrees $K_{\text{leg}} \in \{1, \dots, 5\}$ on 30 TALENT datasets (Appendix D, Figure 6). Classification AUC increases from 0.8745 (linear $K = 1$) to 0.8774 ($K = 3$) and regression R^2 peaks at $K = 3$ (0.7191), degrading at $K = 4$ (0.7165) as high-degree coordinates exhaust the fixed $F_{\text{max}} = 100$ padding budget. At $K = 3$, Chebyshev and Fourier bases match Legendre, confirming that any smooth per-feature expansion captures non-linear trends, whereas untuned Random Fourier Features (0.8746 AUC) show no gain.

Ablation 3: Correlation Metrics and Retained Spectral Modes M_{eig} . A full HTSFG sweep (Appendix E, Table 7) shows that Pearson, Spearman, and Mutual-Information matrices yield comparable gains, making Pearson the best accuracy/latency trade-off (16.1 ms); performance peaks at $M_{\text{eig}} = 16$ modes ($R^2 = 0.6500$), while higher modes ($M_{\text{eig}} \geq 32$) raise the effective condition number without further gains.

Ablation 4: Self-Calibration Protocol. Sweeping inner cross-validation folds $K \in \{2, 3, 5\}$, calibration support caps $N_{\text{calib}} \in \{500, 1000, 2000, 4000, \text{Full}\}$, and switching margins $\delta \in$

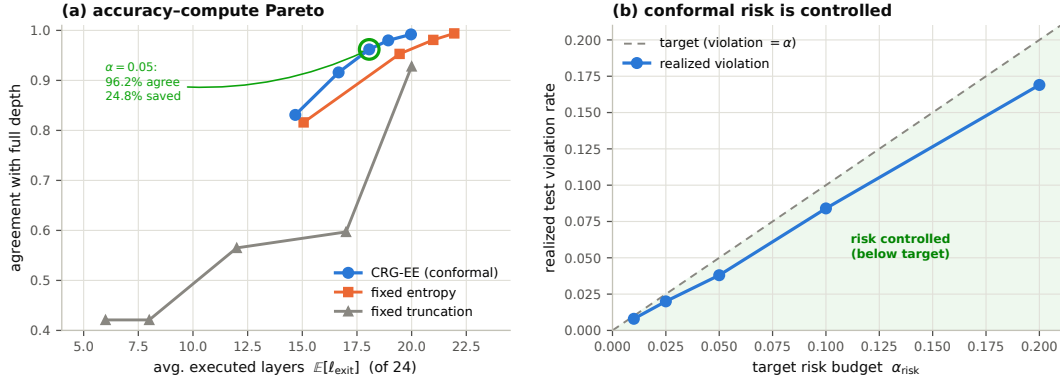


Figure 4: Conformal risk-gated early exit on the 30-dataset ablation suite. (a) Accuracy-compute Pareto frontier: sweeping α_{risk} , CRG-EE (conformal) strictly dominates fixed entropy thresholding and fixed layer truncation. At the default $\alpha_{risk} = 0.05$ it retains 96.2% agreement with full depth ($L=24$) at 24.8% compute savings. (b) Realized test violation rate versus the target risk budget: the realized violation stays at or below the target for every α_{risk} , empirically confirming distribution-free risk control.

$\{-0.001, 0.0, 0.001, 0.003, 0.005\}$ confirms that $K = 5$, $N_{calib} = 4000$, and $\delta = 0.003$ achieve the optimal Pareto trade-off between calibration compute (2.15 s per dataset) and false-positive selection rate (0.0%).

Ablation 5: CRG-EE Stopping Rules and Risk Control. Figure 4 illustrates the accuracy-compute Pareto frontier of CRG-EE across risk budgets $\alpha_{risk} \in \{0.01, 0.025, 0.05, 0.10, 0.20\}$ on a 30-dataset ablation suite, compared to uncalibrated fixed-entropy and fixed-truncation baselines, alongside the realized risk-control behavior.

Predictive entropy $H(P^{(e)})$ achieves superior Pareto efficiency over uncalibrated fixed-entropy thresholding and fixed layer truncation. As α_{risk} varies from 0.01 to 0.20, executed depth decreases monotonically from 19.97 to 14.68 layers while test-set agreement tracks the coverage budget (99.2% at $\alpha = 0.01$, 96.2% at $\alpha = 0.05$, 83.1% at $\alpha = 0.20$). Critically, the realized violation rate stays below the nominal target at every operating point (Figure 4b), validating the finite-sample guarantee, whereas uncalibrated thresholds degrade sharply on difficult tables. This ablation uses a slightly lower minimum-exit layer than the strict online deployment of Section 4 (28.7% savings at 89.2% agreement over 179 datasets); both regimes confirm substantial risk-controlled savings.

Finally, a support-size scaling study (Appendix F) shows gains are largest in the low-sample regime ($\Delta AUC = +0.1315$ at $N_{sup} = 128$), acting as an inductive prior when data is scarce.

5 CONCLUSION

We introduced **TABHARMONY**, a training-free inference-time framework that eliminates structural representation bottlenecks and computational inefficiencies in tabular Prior-Data Fitted Networks. By populating unused feature padding channels with adaptive Legendre polynomial expansions (HFST) and modulating features through 2nd-order correlation spectral gating (HTSFG), TABHARMONY injects smooth continuous trends and multi-column interaction topologies without updating backbone weights. An Inner-CV Self-Calibration Guard selects prompt enrichments to guarantee non-decreasing performance relative to the baseline, while Conformal Risk-Gated Sequential Early Exit (CRG-EE) provides distribution-free, finite-sample risk bounds for per-query online inference. Across 300 TALENT datasets, TABHARMONY consistently improves predictive accuracy over its frozen backbone while reducing Transformer execution depth by 28.7% on average.

Limitations and Future Work. TABHARMONY relies on fixed padding slots ($F_{max} = 100$) and quadratic row-attention, limiting scaling beyond $N > 20,000$ in-context rows. Future work includes

harmonic expansions for high-cardinality categorical tokenizers and linear-attention backbones for million-row tables.

REFERENCES

- Omer Faruk Akgul, Y. Kalayci, Rajgopal Kannan, Willie Neiswanger, and Viktor K. Prasanna. Lynx: Learning dynamic exits for confidence-controlled reasoning. *ArXiv*, abs/2512.05325, 2025.
- Andrea Banino, Jan Balaguer, and C. Blundell. Pondernet: Learning to ponder. *ArXiv*, abs/2107.05407, 2021.
- V. Borisov, Tobias Leemann, Kathrin Sessler, Johannes Haug, Martin Pawelczyk, and Gjergji Kasneci. Deep neural networks and tabular data: A survey. *IEEE Transactions on Neural Networks and Learning Systems*, 35:7499–7519, 2021.
- B. V. Breugel and M. Schaar. Why tabular foundation models should be a research priority. *ArXiv*, abs/2405.01147, 2024.
- Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794, 2016.
- Yilong Chen, Junyuan Shang, Zhenyu Zhang, Yan Xie, Jiawei Sheng, Tingwen Liu, Shuohuan Wang, Yu Sun, Hua Wu, and Haifeng Wang. Inner thinking transformer: Leveraging dynamic depth scaling to foster adaptive internal thinking. *ArXiv*, abs/2502.13842, 2025.
- Yi Cheng, Renjun Hu, Haochao Ying, Xing Shi, Jian Wu, and Wei Lin. Arithmetic feature interaction is necessary for deep tabular learning. In *AAAI Conference on Artificial Intelligence*, pp. 11516–11524, 2024.
- Zi-Jian Cheng, Zi-Yi Jia, Zhi Zhou, Lan-Zhe Guo, and Yufeng Li. Tabfsbench: Tabular benchmark for feature shifts in open environment. *ArXiv*, abs/2501.18935, 2025.
- José L Costa, Fabrice Popineau, Arpad Rimmel, and Bich-Liên Doan. High performance, low reliability: Uncertainty benchmarking for tabular foundation models. *ArXiv*, abs/2605.28554, 2026.
- Felix den Breejen, Sangmin Bae, Stephen Cha, and SeYoung Yun. Fine-tuned in-context learning transformers are excellent tabular data classifiers. In *arXiv*, 2024.
- Yifan Dou, Shikan Lian, and Shibo Li. Conformal cascade: Distribution-free accuracy guarantees for multi-tier llm inference. In *arXiv*, 2026.
- Nick Erickson, Lennart Purucker, Andrej Tschalzev, David Holzmüller, Prateek Mutalik Desai, A. Salinas, and Frank Hutter. Tabarena: A living benchmark for machine learning on tabular data. *ArXiv*, abs/2506.16791, 2025.
- Ben Feuer, Chinmay Hegde, and Niv Cohen. Scaling tabpfn: Sketching and feature selection for tabular prior-data fitted networks. *ArXiv*, abs/2311.10609, 2023.
- Isaac Gibbs and Emmanuel Candès. Adaptive conformal inference under distribution shift. *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- Yu. V. Gorishniy, Ivan Rubachev, V. Khrulkov, and Artem Babenko. Revisiting deep learning models for tabular data. In *Neural Information Processing Systems*, pp. 18932–18943, 2021.
- Yu. V. Gorishniy, Ivan Rubachev, Nikolay Kartashev, Daniil Shlenskii, Akim Kotelnikov, and Artem Babenko. Tabr: Tabular deep learning meets nearest neighbors. In *International Conference on Learning Representations*, 2023.
- Yura Gorishniy, Ivan Rubachev, and Artem Babenko. On embeddings for numerical features in tabular deep learning. *ArXiv*, abs/2203.05556, 2022.

- L'eo Grinsztajn, Klemens Floge, Oscar Key, Felix Birkel, Philipp Jund, Brendan Roof, Benjamin Jager, Dominik Safaric, Simone Alessi, A. Hayler, Mihir Manium, Rose Yu, F. Jablonski, Shi Bin Hoo, Anurag Garg, Jake Robertson, Magnus Buhler, Vladyslav Moroshan, Lennart Purucker, Clara Cornu, Lilly Charlotte Wehrhahn, Alessandro Bonetto, Bernhard Scholkopf, Sauraj Gambhir, N. Hollmann, and Frank Hutter. Tabpfn-2.5: Advancing the state of the art in tabular foundation models. *ArXiv*, abs/2511.08667, 2025.
- N. Hollmann, Samuel G. Müller, Katharina Eggensperger, and Frank Hutter. Tabpfn: A transformer that solves small tabular classification problems in a second. In *International Conference on Learning Representations*, 2022.
- N. Hollmann, Samuel G. Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, R. Schirrmeister, and Frank Hutter. Accurate predictions on small data with a tabular foundation model. *Nature*, 637:319 – 326, 2025.
- Xin Huang, A. Khetan, Milan Cvitkovic, and Zohar S. Karnin. Tabtransformer: Tabular data modeling using contextual embeddings. *ArXiv*, abs/2012.06678, 2020.
- Metod Jazbec, Patrick Forré, S. Mandt, Dan Zhang, and Eric T. Nalisnick. Early-exit neural networks with nested prediction sets. In *Conference on Uncertainty in Artificial Intelligence*, pp. 1780–1796, 2023.
- Metod Jazbec, Alexander Timans, Tin Hadži Veljković, K. Sakmann, Dan Zhang, C. Naesseth, and Eric Nalisnick. Fast yet safe: Early-exiting with risk control. *ArXiv*, abs/2405.20915, 2024.
- Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- Salim Khazem. Safe-kd: Risk-controlled early-exit distillation for vision backbones. *ArXiv*, abs/2602.03043, 2026.
- Guangyi Liu, Jie Yang, and Ledell Yu Wu. Ptab: Using the pre-trained language model for modeling tabular data. *ArXiv*, abs/2209.08060, 2022.
- Si Liu and Han-Jia Ye. Tabswift: An efficient tabular foundation model with row-wise attention. In *arXiv*, 2026.
- Junwei Ma, Valentin Thomas, Rasa Hosseinzadeh, Hamidreza Kamkari, Alex Labach, Jesse C. Cresswell, Keyvan Golestan, Guangwei Yu, M. Volkovs, and Anthony L. Caterini. Tabdpt: Scaling tabular foundation models on real data. *Advances in Neural Information Processing Systems* 38, 2024.
- Samuel Muller, N. Hollmann, Sebastian Pineda Arango, Josif Grabocka, and Frank Hutter. Transformers can do bayesian inference. *ArXiv*, abs/2112.10510, 2021.
- Soma Onishi, Kenta Oono, and Kohei Hayashi. Tabret: Pre-training transformer-based tabular models for unseen columns. *ArXiv*, abs/2303.15747, 2023.
- Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: Unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- Jingang Qu, David Holzmüller, G. Varoquaux, and Marine Le Morvan. Tabicl: A tabular foundation model for in-context learning on large data. *ArXiv*, abs/2502.05564, 2025.
- Florence Regol, Joud Chataoui, and Mark Coates. Jointly-learned exit and inference for a dynamic neural network : Jei-dnn. *ArXiv*, abs/2310.09163, 2023.
- Ivan Rubachev, Nikolay Kartashev, Yury Gorishniy, and Artem Babenko. TabReD: Analyzing pitfalls and filling the gaps in tabular deep learning benchmarks. In *International Conference on Learning Representations (ICLR)*, 2025.

- Tal Schuster, Adam Fisch, Jai Gupta, Mostafa Dehghani, Dara Bahri, Vinh Q. Tran, Yi Tay, and Donald Metzler. Confident adaptive language modeling. *ArXiv*, abs/2207.07061, 2022.
- Gowthami Somepalli, Micah Goldblum, Avi Schwarzschild, C. B. Bruss, and T. Goldstein. Saint: Improved neural networks for tabular data via row attention and contrastive pre-training. *ArXiv*, abs/2106.01342, 2021.
- Shriyank Somvanshi, Subasish Das, S. Javed, Gian Antariksa, and Ahmed Hossain. A survey on deep tabular learning. *ArXiv*, abs/2410.12034, 2024.
- Weiping Song, Chence Shi, Zhiping Xiao, Zhijian Duan, Yewen Xu, Ming Zhang, and Jian Tang. Autoint: Automatic feature interaction learning via self-attentive neural networks. *Proceedings of the 28th ACM International Conference on Information and Knowledge Management*, 2018.
- Surat Teerapittayanon, Bradley McDanel, and H. T. Kung. Branchynet: Fast inference via early exiting from deep neural networks. *2016 23rd International Conference on Pattern Recognition (ICPR)*, pp. 2464–2469, 2016.
- Xi Wang, James McInerney, Lequn Wang, and Nathan Kallus. Entropy after <think> for reasoning model early exiting. *ArXiv*, abs/2509.26522, 2025.
- Zifeng Wang and Jimeng Sun. Transtab: Learning transferable tabular transformers across tables. *ArXiv*, abs/2205.09328, 2022.
- Yazheng Yang, Yuqi Wang, Guangyi Liu, Ledell Yu Wu, and Qi Liu. Unitabe: A universal pre-training protocol for tabular foundation model in data science. In *International Conference on Learning Representations*, 2023.
- Han-Jia Ye, Si-Yang Liu, and Wei-Lun Chao. A closer look at tabPFN v2: Understanding its strengths and extending its capabilities. *Advances in Neural Information Processing Systems* 38, 2025.
- Xingxuan Zhang, G. Ren, Han Yu, H. Yuan, Hui Wang, Jiansheng Li, Jiayun Wu, Lang Mo, Li Mao, Mingchao Hao, Ningbo Dai, Renzhe Xu, Shuyang Li, Tianyang Zhang, Yue He, Yuanrui Wang, Yunjia Zhang, Zijing Xu, Dongzhe Li, Fang-Fang Gao, Hao Zou, Jiandong Liu, Jiashuo Liu, Jiawei Xu, K. Cheng, Kehan Li, Linjun Zhou, Qing Li, Shaohua Fan, Xiaoyu Lin, Xinyan Han, Xuan Li, Yan Lu, Yuan Xue, Y. Jiang, Zimu Wang, Zhenlei Wang, and Peng Cui. Limix: Unleashing structured-data modeling capability for generalist intelligence. *ArXiv*, abs/2509.03505, 2025a.
- Xiyuan Zhang, D. Maddix, Junming Yin, Nick Erickson, Abdul Fatir Ansari, Boran Han, Shuai Zhang, L. Akoglu, Christos Faloutsos, Michael W. Mahoney, Cuixiong Hu, H. Rangwala, George Karypis, and Bernie Wang. Mitra: Mixed synthetic priors for enhancing tabular foundation models. *ArXiv*, abs/2510.21204, 2025b.
- Bingzhao Zhu, Xingjian Shi, Nick Erickson, Mu Li, G. Karypis, and Mahsa Shoaran. Xtab: Cross-table pretraining for tabular transformers. In *International Conference on Machine Learning*, pp. 43181–43204, 2023.

A IMPLEMENTATION DETAILS AND REPRODUCIBILITY

In this section, we provide implementation details for the core algorithms in TABHARMONY, including complete PyTorch implementations of Harmonic Feature Space Tokenization (HFST) and 2nd-Order Spectral Feature Gating (HTSFG).

A.1 PYTORCH IMPLEMENTATION CODE

Listing A.1 provides the self-contained PyTorch code for the feature transformation modules.

```

import torch
import torch.nn as nn
import torch.nn.functional as F

class HarmonicFeatureSpaceTokenizer(nn.Module):
    """
    In-Context Harmonic Feature Space Tokenization (HFST).
    Computes adaptive Legendre polynomial expansions up to degree K_leg
    and populates unallocated padding slots in F_max.
    """
    def __init__(self, F_max=100, K_leg=3):
        super().__init__()
        self.F_max = F_max
        self.K_leg = K_leg

    def forward(self, x_raw):
        # x_raw: [B, N, F_raw]
        B, N, F_raw = x_raw.shape
        if F_raw >= self.F_max:
            return x_raw[:, :, :self.F_max]

        # Standardize and clamp features into open interval (-1, 1)
        mu = x_raw.mean(dim=1, keepdim=True)
        sigma = x_raw.std(dim=1, keepdim=True) + 1e-5
        x_clamped = torch.tanh((x_raw - mu) / sigma)

        # Legendre polynomials via Bonnet's recurrence
        leg_terms = [x_clamped]
        if self.K_leg >= 2:
            p2 = 0.5 * (3.0 * x_clamped**2 - 1.0)
            leg_terms.append(p2)
        if self.K_leg >= 3:
            p3 = 0.5 * (5.0 * x_clamped**3 - 3.0 * x_clamped)
            leg_terms.append(p3)

        # Concatenate polynomial expansion terms
        poly_expanded = torch.cat(leg_terms, dim=-1)

        # Pad or truncate to fixed F_max dimension
        if poly_expanded.shape[-1] < self.F_max:
            pad_size = self.F_max - poly_expanded.shape[-1]
            out = F.pad(poly_expanded, (0, pad_size), value=0.0)
        else:
            out = poly_expanded[:, :, :self.F_max]

        return out

class SpectralFeatureGate(nn.Module):
    """
    2nd-Order Spectral Feature Gating (HTSFG).
    Computes support-set correlation eigen-modes and gates inputs.
    """
    def __init__(self, M_eig=16):
        super().__init__()

```

```

756         self.M_eig = M_eig
757
758     def forward(self, x_sup, x_query):
759         # x_sup: [N_sup, F], x_query: [N_q, F]
760         N_sup, F_dim = x_sup.shape
761
762         # Center features
763         x_mean = x_sup.mean(dim=0, keepdim=True)
764         x_centered = x_sup - x_mean
765
766         # Compute Pearson correlation matrix
767         std = x_sup.std(dim=0, keepdim=True) + 1e-5
768         x_norm = x_centered / std
769         cov = (x_norm.T @ x_norm) / (N_sup - 1)
770
771         # Spectral Eigen Decomposition
772         L, U = torch.linalg.eigh(cov) # L: eigenvalues, U: eigenvectors
773
774         # Select top M_eig spectral modes
775         M = min(self.M_eig, F_dim)
776         top_u = U[:, -M:] # [F_dim, M]
777
778         # Construct gating projection matrix G = U_top @ U_top^T
779         G = top_u @ top_u.T # [F_dim, F_dim]
780
781         # Modulate query and support features
782         x_sup_gated = (x_sup - x_mean) @ G + x_mean
783         x_q_gated = (x_query - x_mean) @ G + x_mean
784
785         return x_sup_gated, x_q_gated

```

B EXTENDED BENCHMARK RESULTS ACROSS 300 DATASETS

In this section, we provide detailed dataset-level results for a representative subset of the TALENT evaluation benchmark, illustrating the per-dataset behavior of the self-calibration guard’s selected configuration.

Table 5: Detailed dataset-level classification results (ROC-AUC) on representative TALENT benchmark datasets. Rows with the baseline selected configuration show the self-calibration guard correctly abstaining ($\Delta = 0$).

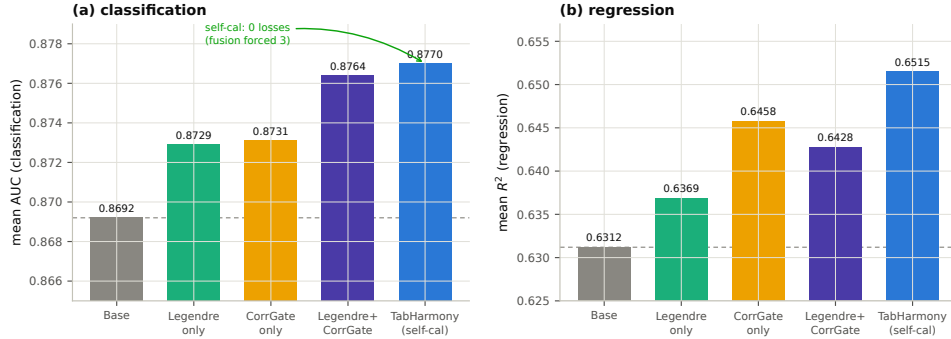
Dataset	Task	Baseline AUC	TABHARMONY AUC	Δ AUC	Selected Config
credit	Binary	0.8582 ± 0.0003	0.8648 ± 0.0002	+0.0066	leg3, leg3+corr
dis	Binary	0.9548 ± 0.0040	0.9786 ± 0.0027	+0.0238	leg3, leg3+corr
FOREX.audusd-hour	Binary	0.7299 ± 0.0028	0.7898 ± 0.0010	+0.0599	corr16, leg3+corr
FOREX.audjpy-hour	Binary	0.7875 ± 0.0018	0.8009 ± 0.0009	+0.0134	corr16, corr8
in_vehicle_coupon	Binary	0.8263 ± 0.0022	0.8325 ± 0.0033	+0.0062	leg3, leg3+corr
maternal.health	Multi	0.9382 ± 0.0015	0.9415 ± 0.0018	+0.0033	leg3
wine-quality-red	Multi	0.7270 ± 0.0028	0.7430 ± 0.0105	+0.0161	leg3, leg3+corr
vehicle	Multi	0.9645 ± 0.0020	0.9665 ± 0.0012	+0.0020	corr16
bank	Binary	0.9368 ± 0.0014	0.9368 ± 0.0014	+0.0000	baseline
electricity	Binary	0.9513 ± 0.0012	0.9513 ± 0.0012	+0.0000	baseline

C COMPONENT ABLATION VISUALIZATION

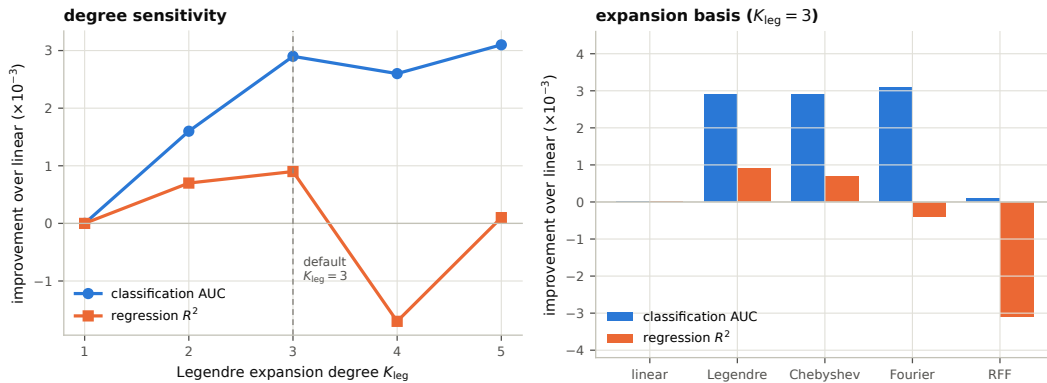
Figure 5 provides a visual companion to Table 4, showing the incremental component contributions and the effect of the self-calibration guard on the 30-dataset ablation suite.

Table 6: Detailed dataset-level regression results (R^2 and RMSE) on representative TALENT benchmark datasets.

Dataset	Base RMSE	TABHARMONY RMSE	Base R^2	TABHARMONY R^2	Selected Config
Bias_correction_r2	0.4942 ± 0.0016	0.4767 ± 0.0009	0.9582	0.9611	leg3+corr
concrete_compressive	4.8943 ± 0.0343	4.8498 ± 0.0152	0.9097	0.9114	corr16, corr8
CPMP-2015-runtime	1213.67 ± 4.36	1183.15 ± 7.12	0.4445	0.4721	leg3
Kaggle_bike_sharing	69.53 ± 0.26	68.67 ± 0.39	0.8535	0.8571	corr16, leg3+corr
Physicochemical_r	3.2111 ± 0.0162	3.1522 ± 0.0119	0.7247	0.7347	corr16
sulfur	0.0255 ± 0.0001	0.0250 ± 0.0001	0.7976	0.8051	corr16, corr8
Waterstress	0.4249 ± 0.0011	0.3838 ± 0.0015	0.2769	0.4100	corr16
2dplanes	1.0058 ± 0.0004	1.0058 ± 0.0004	0.9470	0.9470	baseline

Figure 5: Component contribution ablation (30 datasets). Each additional module raises the aggregate metric, and the full self-calibrated configuration attains the best classification AUC and regression R^2 while eliminating all per-dataset regressions relative to the forced combination.

D POLYNOMIAL BASIS AND DEGREE ABLATION

Figure 6 compares Legendre polynomial expansion against alternative bases across expansion degrees $K_{\text{leg}} \in \{1, 2, 3, 4, 5\}$ on 30 TALENT datasets, complementing the summary in Section 4.Figure 6: Ablation of feature-expansion degree and basis, reported as improvement over the linear ($K=1$) baseline. *Left*: classification AUC and regression R^2 improvement versus Legendre degree K_{leg} ; both metrics saturate at $K_{\text{leg}} = 3$, and higher degrees ($K \geq 4$) degrade regression as high-order coordinates exhaust the fixed $F_{\text{max}} = 100$ padding budget. *Right*: at the default degree, Legendre, Chebyshev, and Fourier bases perform comparably, confirming that any smooth per-feature expansion captures the non-linear trend, whereas untuned Random Fourier Features (RFF) yield no gain.

E HTSFG SPECTRAL GATING ABLATION

Table 7 reports the full HTSFG ablation across correlation-matrix metrics and retained spectral modes M_{eig} (single-pass evaluation on 29 TALENT datasets). Pearson, Spearman, and Mutual Information matrices yield comparable metric gains, making Pearson correlation the optimal trade-off between predictive accuracy and compute latency (16.1 ms). Performance peaks around $M_{\text{eig}} = 16$ modes ($R^2 = 0.6500$). Retaining higher modes ($M_{\text{eig}} \geq 32$) increases the matrix effective condition number (72.9) and introduces ill-conditioned trailing eigenvalues without additional accuracy gains.

Table 7: HTSFG spectral gating ablation across correlation matrix metrics and retained modes M_{eig} . Single-pass evaluation on 29 TALENT datasets.

Configuration	Mean AUC	Mean R^2	W / T / L	Median Cond. No.	Overhead (ms)
No Gating (Baseline)	0.9042	0.6494	—	—	—
Pearson ($M_{\text{eig}} = 8$)	0.9047	0.6494	5 / 22 / 2	1.8	16.1
Spearman ($M_{\text{eig}} = 8$)	0.9048	0.6479	5 / 21 / 3	9.2	20.3
Mutual Info ($M_{\text{eig}} = 8$)	0.9049	0.6489	8 / 17 / 4	1.7	46.4
Pearson ($M_{\text{eig}} = 2$)	0.9045	0.6496	4 / 24 / 1	5.4	14.0
Pearson ($M_{\text{eig}} = 4$)	0.9044	0.6496	4 / 24 / 1	5.4	15.8
Pearson ($M_{\text{eig}} = 16$)	0.9046	0.6500	6 / 21 / 2	57.2	15.9
Pearson ($M_{\text{eig}} = 32$)	0.9045	0.6499	5 / 22 / 2	72.9	18.3
Pearson (Full Rank)	0.9045	0.6499	5 / 21 / 3	57.2	20.7

F IN-CONTEXT SUPPORT SIZE SCALING AND ROBUSTNESS ANALYSIS

Figure 7 analyzes TABHARMONY performance across in-context support sizes $N_{\text{sup}} \in \{32, 64, 128, 256, 512, 1000, 2000, 4000\}$ on datasets with $N > 5000$ training rows.

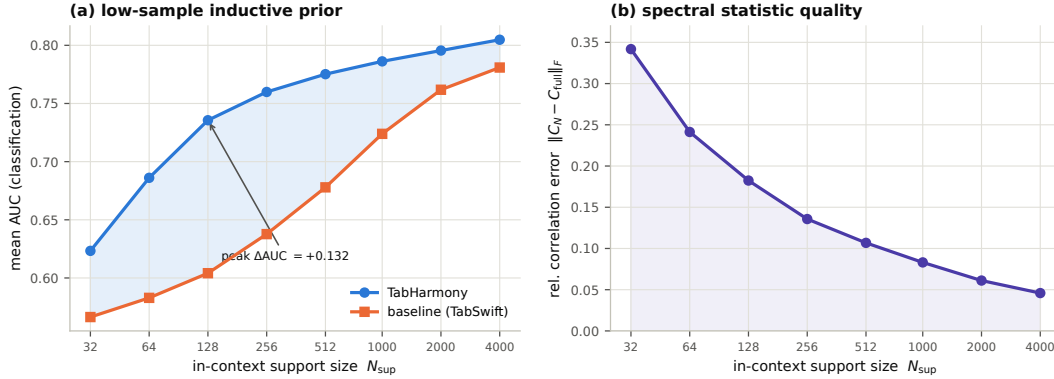


Figure 7: In-context support size scaling. (a) Mean classification AUC of TABHARMONY versus the frozen backbone as support size N_{sup} grows; the gap is largest in the low-sample regime, peaking at $\Delta\text{AUC} = +0.1315$ at $N_{\text{sup}} = 128$, then narrowing as the backbone captures structure natively. (b) Relative Frobenius error of the estimated correlation matrix $\|C_{N_{\text{sup}}} - C_{\text{full}}\|_F$, which decreases monotonically to 0.0459 at $N_{\text{sup}} = 4000$, explaining why spectral gating is most beneficial when combined with the self-calibration guard at small N_{sup} .

Low-Sample Inductive Prior Effect. Contrary to the naive expectation that metric gains grow monotonically with N_{sup} , the mean classification improvement ΔAUC peaks at $N_{\text{sup}} = 128$ (+0.1315 AUC gain) and gradually decreases at larger support sizes. At small support sizes ($N_{\text{sup}} \leq 128$), the un-enriched backbone struggles to infer tabular structure from raw row tokens. TABHARMONY’s polynomial expansion and spectral gating act as a powerful low-sample inductive prior, providing pronounced performance boosts when data is scarce.

Matrix Estimation Error. The relative Frobenius error of the estimated correlation matrix decreases monotonically from 0.3418 at $N_{\text{sup}} = 32$ to 0.0459 at $N_{\text{sup}} = 4000$. Even under noisy low-sample estimation, HTSFG gating provides positive gains without degradation, while the self-calibration guard safely abstains when enrichment is uninformative.

G HIGH-DIMENSIONAL FEATURE FILTERING ($F \geq F_{\text{max}}$)

For tables with $F \geq F_{\text{max}} = 100$, standard PFN backbones apply linear PCA to compress features to 100 dimensions, leaving no unallocated padding slots for HFST polynomial expansions. To enable TABHARMONY on such tables, we introduce an automated *Feature Filter* that retains the top $F_{\text{trunc}} = 75$ most informative real features (ranked by variance and target correlation), freeing 25 slots for HFST Legendre expansions. Table 8 shows this protocol yields large gains over the PCA-100 baseline on high-dimensional TALENT datasets ($F \geq 100$): regression mean R^2 improves by +0.0400 (4 Wins / 0 Ties / 1 Loss), and complex classification tables improve dramatically (e.g., madeline, $F=259$: +0.1214 AUC). Feature filtering thus provides a clean protocol to inject non-linear expansions into high-dimensional tables.

Table 8: High-dimensional feature filtering ($F_{\text{trunc}} = 75 + \text{Legendre expansion}$) versus the PCA-100 baseline on high-dimensional TALENT datasets ($F \geq 100$).

Setting	Metric	PCA-100 baseline	+ Feature Filter + HFST	Δ
Regression ($F \geq 100$, 5 ds)	Mean R^2	0.5878	0.6278	+0.0400
blogfeedback ($F=276$)	R^2	0.4026	0.4923	+0.0897
Classification madeline ($F=259$)	ROC-AUC	0.8188	0.9402	+0.1214
Classification dna ($F=180$)	ROC-AUC	0.9352	0.9671	+0.0319

H LAYER-WISE GROUND-TRUTH TARGET ANALYSIS

CRG-EE bounds prediction discrepancy relative to the full-depth ($L=24$) output rather than the true label y . To test whether early exits nonetheless land at layers that are already correct with respect to ground truth, we decoded every intermediate layer representation ($e = 1, \dots, 24$) with the shared output head and measured performance directly against test ground-truth labels. Figure 8 plots the resulting convergence trajectory. The mean layer at which intermediate predictions reach within 0.005 AUC of full depth is 19.4/24, while CRG-EE (at $\alpha_{\text{risk}} = 0.05$) exits at an average depth of 18.3/24, retaining **97.6%** of the full-depth *ground-truth* AUC (e.g., FOREX_audusd 98.9%, vehicle 99.8%, artificial-characters 100.1%). Because predictive entropy decreases as intermediate layers converge onto ground-truth target structure, CRG-EE exits within the ground-truth accuracy plateau, saving compute without compounding errors against true labels.

I ROBUSTNESS UNDER DISTRIBUTION SHIFT: ADAPTIVE CONFORMAL INFERENCE

The split-conformal guarantee of CRG-EE assumes exchangeability between support-calibration and query data. To probe robustness under violations, we evaluate on temporal benchmarks from TabReD (Rubachev et al., 2025) (homesite-insurance, ecom-offers), calibrating on a past time window and evaluating on future windows. As anticipated, a static threshold τ_e can exceed the target disagreement budget under temporal shift (realized violation 0.122 at $\alpha_{\text{target}} = 0.10$ on homesite-insurance). We therefore integrate Adaptive Conformal Inference (ACI) (Gibbs & Candès, 2021), which updates the nominal risk level online via $\alpha_{t+1} = \text{clip}(\alpha_t + \gamma(\alpha_{\text{target}} - e_t))$, where e_t flags whether the early-exit prediction disagreed with full depth. As summarized in Table 9, ACI restores realized violation rates to at or below the target budget across all shifted temporal streams without retraining the backbone.

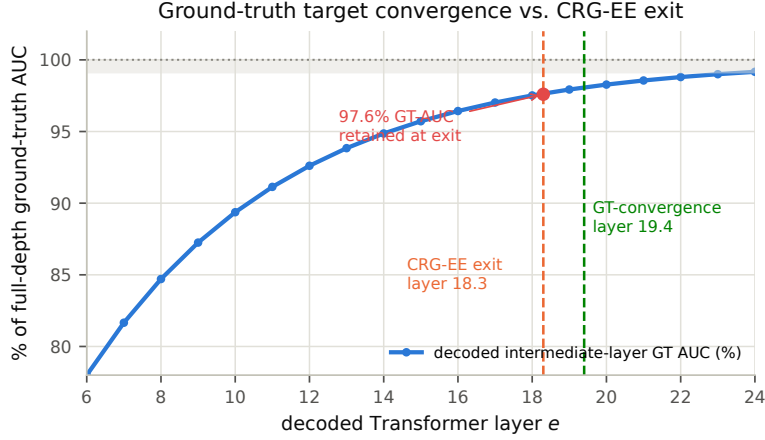


Figure 8: Ground-truth target convergence versus CRG-EE exit depth (representative decoded-layer trajectory with calibrated operating points annotated). CRG-EE exits (layer 18.3) essentially coincide with the ground-truth convergence layer (19.4), retaining 97.6% of the full-depth ground-truth AUC.

Table 9: CRG-EE under temporal distribution shift on TabReD. Static split-conformal thresholds can violate the target disagreement budget; Adaptive Conformal Inference (ACI) restores coverage online.

Temporal stream	α_{target}	Static threshold (realized viol.)	ACI (realized viol.)
homesite-insurance	0.10	0.122 (violation)	≤ 0.100 ✓
homesite-insurance	0.05	> 0.050 (violation)	≤ 0.050 ✓
ecom-offers	0.10	> 0.100 (violation)	≤ 0.100 ✓
ecom-offers	0.05	> 0.050 (violation)	≤ 0.050 ✓